
Session 2 — Linear algebra II

Exercise sheet

Determinant, rank, eigenvalues, symmetric matrices, PCA and the SVD. The recurring picture is the same data table as last time — rows are participants, columns are measures — and the question of the day is: how many directions does it really have? Show your reasoning; most answers are short. Nothing here repeats a question already answered on the slides: the slides give the method, this sheet asks you to run it.

Exercise 1 Determinants and inverses.

- (a) Compute $\det \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$ and the inverse.
- (b) Compute $\det \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ and the inverse.
- (c) Compute $\det \begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix}$. Is the matrix invertible? What is its rank?
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Exercise 2

- (a) For which k is $\begin{pmatrix} 2 & 3 \\ 4 & k \end{pmatrix}$ non-invertible, and what happens to its rank there? Write the critical matrix as an outer product.
- (b) For which k is $\begin{pmatrix} 1 & 4 \\ 3 & k \end{pmatrix}$ non-invertible?
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Exercise 3 The determinant as a signed area factor.

- (a) $\det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1$. What do the size and the sign each mean geometrically?
- (b) With $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, check that $\det(AB) = \det A \det B$.
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Exercise 4 Rank.

- (a) What is the rank of $\begin{pmatrix} 2 & 6 \\ 3 & 9 \end{pmatrix}$? Does it have an inverse, and on what line do all its outputs lie?

(b) What is the rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$?

(c) Give a 2×2 matrix of rank 0, one of rank 1, and one of rank 2.

Exercise 5 *A table has a redundant column whenever one column is a linear combination of the others. The rank is then smaller than the number of columns, and no analysis can recover a unique weight for each column.*

A study records, for each participant, the score on item 1, the score on item 2, and the *difference* item 1 – item 2:

$$X = \begin{pmatrix} 4 & 9 & -5 \\ 6 & 7 & -1 \\ 5 & 8 & -3 \\ 9 & 4 & 5 \\ 7 & 6 & 1 \end{pmatrix}.$$

(a) Find a combination of the three columns that gives the zero vector, and deduce the rank of X . Justify why the rank is not smaller.

(b) The weights $\mathbf{w} = (1, -1, 0)$ and $\mathbf{w}' = (0, 0, 1)$ are different. Show that $X\mathbf{w} = X\mathbf{w}'$, and give a third weight vector with the same product.

(c) In one sentence: what does that mean for a regression using all three columns as predictors?

Exercise 6 *An outer product $\mathbf{m}\mathbf{n}^\top$ is the simplest non-trivial matrix there is. Read as a table it says: everybody has the same profile $\mathbf{n}$, only the overall level m_i differs.*

Compute the outer product $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix}$, then its determinant and its rank.

Exercise 7

(a) A 200×20 table of responses is exactly $X = \mathbf{m}\mathbf{n}^\top$, with $\mathbf{m} \in \mathbb{R}^{200}$ and $\mathbf{n} \in \mathbb{R}^{20}$. What is its rank? How many numbers would you need to store instead of the 4000 entries?

(b) Now $X = \mathbf{m}_1\mathbf{n}_1^\top + \mathbf{m}_2\mathbf{n}_2^\top$, with $\mathbf{m}_1, \mathbf{m}_2$ independent and $\mathbf{n}_1, \mathbf{n}_2$ independent. Give the rank, and describe in words what the table looks like.

Exercise 8

Eigenvectors by inspection.

(a) Verify that $(2, 1)$ is an eigenvector of $\begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$ and give its eigenvalue. Is $(4, 2)$ also an eigenvector?

(b) Give the eigenvalues of $\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$ and of $\begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$, with their eigenvectors.

Exercise 9 Use the trace T and determinant D , and the characteristic equation $\lambda^2 - T\lambda + D = 0$.

- (a) $\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$: give T , D , the eigenvalues and the eigenvectors.
- (b) $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$: give T , D and the eigenvalues.

Exercise 10 Find the eigenvalues and eigenvectors of:

- (a) $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$;
- (b) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$;
- (c) $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$;
- (d) $\begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$.

Exercise 11 Complex eigenvalues.

- (a) Find the eigenvalues of $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ and of $\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$. Real or complex, and what do they mean geometrically?
- (b) For which real a does $\begin{pmatrix} 1 & a \\ -a & 1 \end{pmatrix}$ have complex eigenvalues?

Exercise 12 *A repeated eigenvalue is often blamed for a matrix failing to be diagonalizable. These two examples show that the blame is misplaced: what matters is how many independent eigenvectors you can find, not how many distinct eigenvalues there are.*

- (a) Show that $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ has $\lambda = 3$ twice, then solve $(A - 3I)\mathbf{v} = \mathbf{0}$ and count the independent eigenvectors. Is it diagonalizable?
- (b) Do the same for $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$. How many independent eigenvectors does it have?
- (c) In one sentence: what do (a) and (b) together say about the claim “a repeated eigenvalue means the matrix cannot be diagonalized”?

Exercise 13 *Applying the same linear rule over and over — a learning rule trial after trial, an iterative algorithm — is governed entirely by the eigenvalues.*

- (a) A matrix has eigenvalues -2 and 0.5 . Apply it many times to a generic vector: which eigendirection does the result line up with, and what does the *sign* of the dominant eigenvalue add to the picture?
- (b) With $A = P\Lambda P^{-1}$ and $\lambda_1 = 1$, $\lambda_2 = \frac{1}{2}$, describe A^{10} : which mode survives?
- (c) Using the eigendecomposition found in Exercise 10(b), compute A^3 for $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, and check it against direct multiplication.

Exercise 14 A 2×2 matrix has eigenvalues -3 and 5 . Give its trace and its determinant, without knowing the matrix.

Exercise 15 Symmetric matrices.

- (a) Check that $\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ is symmetric, find its eigenvalues, and verify that its two eigenvectors are orthogonal.
- (b) Do the same for $\begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$.
- (c) Give a 2×2 *real* matrix whose eigenvalues are complex, and then explain — without computing anything else — why that matrix cannot possibly be symmetric.

Exercise 16 *For centred data, $C = \frac{1}{n}X^T X$ is the covariance matrix: variances on the diagonal, co-variation off it. It is symmetric and positive semi-definite, its eigenvectors are the principal components, and the matching eigenvalues are the variances along them.*

- (a) A covariance matrix has eigenvalues 8 and 0.5 . Along which eigenvector does the data mostly spread, and what fraction of the total variance does that direction carry?
- (b) For $C = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$, give the two principal components in words (the two measures are the scores on two similar tasks), and the fraction of variance along the first. Check that $\text{tr } C$ is the sum of the eigenvalues.

Exercise 17 Which of these could be a covariance matrix? Justify each answer with the eigenvalues.

- (a) $\begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$ — and which axis is then the first principal component?
- (b) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ — and what is its rank?
- (c) $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ (you found its eigenvalues in Exercise 10(b));
- (d) $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

Exercise 18 Four participants, two measures, already centred: $(1, 1)$, $(-1, -1)$, $(2, 2)$, $(-2, -2)$.

- (a) Compute $C = \frac{1}{4} \sum_i \mathbf{x}_i \mathbf{x}_i^\top$.
- (b) Give its eigenvalues and the first principal component.
- (c) What does the second eigenvalue tell you about the two measures?

Exercise 19 Now the four centred points are $(2, 0)$, $(0, 2)$, $(-2, 0)$, $(0, -2)$. Compute C and say what is special about its principal components.

Exercise 20 A data table is $n \times p$ and almost never square, so it has no eigenvalues at all. The singular value decomposition $X = U\Sigma V^\top$ exists for every matrix: U and V are orthogonal, Σ is diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

- (a) A centred table X is 200×6 . Give the dimensions of U , Σ and V , and say how many singular values there are.
- (b) Which of the three factors holds the directions in *measure* space (i.e. the principal components)? Which holds one row per participant?
- (c) Why can we not simply take eigenvalues of X instead?

Exercise 21 In the lecture we expanded $X^\top X$ and found V . Here you do the other side of the same computation — the one that explains what U is.

Start from $X = U\Sigma V^\top$ with $U^\top U = I$ and $V^\top V = I$.

- (a) Show that $XX^\top = U(\Sigma\Sigma^\top)U^\top$. Of which matrix are the columns of U therefore the eigenvectors, and what does that matrix compare?
- (b) $\Sigma^\top\Sigma$ and $\Sigma\Sigma^\top$ do not even have the same size. Explain why $X^\top X$ and $XX^\top$ nevertheless have exactly the same *nonzero* eigenvalues.

- (c) X is 200×6 . Give the sizes of $X^T X$ and of XX^T , and say how many eigenvalues of XX^T must be zero.
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Exercise 22 Four participants, two measures, already centred:

$$X = \begin{pmatrix} 3 & 1 \\ 1 & 3 \\ -3 & -1 \\ -1 & -3 \end{pmatrix}.$$

- (a) Compute $X^T X$, then its eigenvalues, then the two singular values of X .
- (b) Give $\mathbf{v}_1$ and $\mathbf{v}_2$, then compute $\mathbf{u}_k = X\mathbf{v}_k/\sigma_k$ for $k = 1, 2$. Check that both have norm 1 and are orthogonal to each other.
- (c) What fraction of the variance does the first component carry?
- (d) Write the best rank-1 approximation $X_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$ explicitly, and give the error $X - X_1$ together with its rank. Check that $X = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T$.
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Exercise 23 A centred table X has singular values $\sigma = (6, 3, 2, 1)$.

- (a) What fraction of the variance is carried by the first component? By the first two?
- (b) Write down the best rank-2 approximation X_2 in terms of the $\mathbf{u}_k$, $\mathbf{v}_k$ and σ_k , and give the rank of the error $X - X_2$.
- (c) A colleague reports “the first two components explain 75% of the variance” using σ rather than σ^2 . Recompute their number and say why it is wrong.
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Exercise 24 Twenty questionnaire items are answered by 500 people. The singular values of the centred table drop sharply after the second one.

- (a) Roughly what rank does the table have, and roughly what rank does it *effectively* have? Are these the same question?
- (b) How many numbers does the rank-2 truncation need, compared with the 10 000 entries of the table?
- (c) In one sentence, say what the truncation is *not* entitled to claim.